

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA5115

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO3: *Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.CO (BL4 , BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 50

TOTAL MARKS: 50

Annexure A

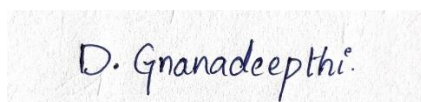
MCQ

Code of Conduct:

I D. Gnana Deepthi(192411123) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink that reads "D. Gnana Deepthi".

Annexure A

MCQ

1. A system uses MD5 for integrity checking. Which attack can most likely compromise it?
 - A. Replay attack
 - B. Collision attack
 - C. Phishing attack
 - D. Brute force on keys
2. Which modification improves security when replacing MD5 in a password system?
 - A. Use plaintext storage
 - B. Use SHA-256 with salting
 - C. Use shorter hash values
 - D. Disable hashing
3. In a digital signature system, what ensures non-repudiation?
 - A. Symmetric key
 - B. Private key of sender
 - C. Public key of receiver
 - D. Hash length
4. Which scenario indicates a failure of a hash function?
 - A. Slow computation
 - B. Same hash for different inputs
 - C. Large output size
 - D. Deterministic output
5. HMAC is preferred over simple hashing because it:
 - A. Reduces computation time
 - B. Uses public keys
 - C. Combines key with hash
 - D. Eliminates encryption
6. In Kerberos, what prevents replay attacks?
 - A. Password reuse
 - B. Timestamp mechanism
 - C. Hash collisions
 - D. Key length
7. Which component in Kerberos issues tickets?
 - A. Client
 - B. Server

- C. KDC
- D. Firewall

8. X.509 certificates primarily provide:
 - A. Data compression
 - B. Authentication
 - C. Routing
 - D. Encryption only
9. What is the main weakness of MD5?
 - A. Slow hashing
 - B. Collision vulnerability
 - C. Large key size
 - D. Complex implementation
10. SHA-256 is preferred over MD5 because it:
 - A. Is faster
 - B. Has shorter output
 - C. Is collision resistant
 - D. Uses symmetric keys
11. In digital signatures, hashing is used to:
 - A. Encrypt message
 - B. Reduce message size
 - C. Generate keys
 - D. Authenticate network
12. Which attack targets MAC without key knowledge?
 - A. Collision attack
 - B. Dictionary attack
 - C. Birthday attack
 - D. Phishing
13. What is required for verifying a digital signature?
 - A. Sender's private key
 - B. Receiver's private key
 - C. Sender's public key
 - D. Shared secret key
14. In ElGamal signature, security depends on:
 - A. Hash length
 - B. Discrete logarithm problem
 - C. RSA problem
 - D. AES encryption

15. Schnorr protocol is mainly used for:
- A. Encryption
 - B. Authentication
 - C. Compression
 - D. Routing
16. Which property ensures hash uniqueness?
- A. Determinism
 - B. Collision resistance
 - C. Speed
 - D. Length
17. A weak MAC allows:
- A. Faster encryption
 - B. Message tampering
 - C. Reduced bandwidth
 - D. Secure authentication
18. Kerberos relies on which cryptographic method?
- A. Asymmetric only
 - B. Symmetric key cryptography
 - C. Hashing only
 - D. Quantum cryptography
19. X.509 certificates include:
- A. Private key
 - B. Public key
 - C. Session key
 - D. Hash key
20. Which attack exploits hash collisions?
- A. MITM
 - B. Birthday attack
 - C. Replay attack
 - D. DoS
21. Digital Signature Standard (DSS) is based on:
- A. AES
 - B. DSA
 - C. DES
 - D. RC4
22. Which ensures integrity in HMAC?
- A. Encryption
 - B. Keyed hashing
 - C. Compression
 - D. Encoding

23. In Kerberos, TGT stands for:
- A. Ticket Granting Ticket
 - B. Trusted Gateway Token
 - C. Temporary Grant Ticket
 - D. Token Generation Table
24. Which protocol uses certificates extensively?
- A. Kerberos
 - B. X.509
 - C. HMAC
 - D. MD5
25. Which property is violated if attacker finds two inputs with same hash?
- A. Pre-image resistance
 - B. Collision resistance
 - C. Determinism
 - D. Efficiency
26. ElGamal signatures are vulnerable if:
- A. Key reused
 - B. Random value reused
 - C. Hash used
 - D. Public key shared
27. Which improves MAC security?
- A. Removing key
 - B. Increasing key randomness
 - C. Using plaintext
 - D. Reducing hash size
28. Which is NOT a hash property?
- A. Deterministic
 - B. Fixed length
 - C. Reversible
 - D. Fast computation
29. Kerberos authentication fails if:
- A. Time not synchronized
 - B. Hash too long
 - C. Key too short
 - D. Network fast
30. X.509 trust depends on:
- A. Hash length
 - B. Certificate authority
 - C. Encryption speed
 - D. MAC value

31. Schnorr signatures are considered efficient because:
- A. Smaller signature size
 - B. Larger keys
 - C. No hashing
 - D. No authentication
32. Which is a MAC algorithm?
- A. SHA-256
 - B. HMAC
 - C. RSA
 - D. ElGamal
33. Digital signatures provide:
- A. Confidentiality only
 - B. Integrity and authentication
 - C. Compression
 - D. Routing
34. Which algorithm is obsolete due to security flaws?
- A. SHA-256
 - B. MD5
 - C. HMAC
 - D. DSS
35. In Kerberos, session keys are used for:
- A. Permanent storage
 - B. Temporary communication
 - C. Hashing
 - D. Compression
36. Which ensures sender authenticity in email?
- A. Hash only
 - B. Digital signature
 - C. Encryption only
 - D. MAC only
37. Which attack is prevented by HMAC?
- A. Replay attack
 - B. Message tampering
 - C. DoS
 - D. Phishing
38. X.509 certificates are validated using:
- A. Symmetric keys
 - B. Public key infrastructure
 - C. Hash only
 - D. MAC

39. Which is true about SHA algorithms?
- A. Reversible
 - B. Collision resistant
 - C. Symmetric
 - D. Encrypted
40. Which parameter is critical in ElGamal security?
- A. Prime number
 - B. Hash length
 - C. MAC key
 - D. Timestamp
41. Which step verifies a digital signature?
- A. Encrypt hash
 - B. Decrypt signature with public key
 - C. Generate MAC
 - D. Compress message
42. Weak hash functions affect:
- A. Bandwidth
 - B. Integrity
 - C. Routing
 - D. Compression
43. Kerberos uses which server?
- A. DNS
 - B. KDC
 - C. HTTP
 - D. FTP
44. Which provides integrity and authenticity together?
- A. Encryption
 - B. MAC
 - C. Routing
 - D. Compression
45. Schnorr protocol relies on:
- A. RSA problem
 - B. Discrete log problem
 - C. AES
 - D. DES
46. Which is vulnerable to collision attacks?
- A. MD5
 - B. SHA-512
 - C. HMAC
 - D. DSS

47. Which improves digital signature security?
- A. Larger key size
 - B. Smaller hash
 - C. No hashing
 - D. Plaintext
48. Which ensures certificate authenticity?
- A. MAC
 - B. CA signature
 - C. Hash only
 - D. Encryption
49. What is the main role of MAC?
- A. Confidentiality
 - B. Authentication
 - C. Compression
 - D. Routing
50. Which combination is most secure for integrity?
- A. MD5 only
 - B. SHA + HMAC
 - C. Plaintext
 - D. Encryption only

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Total points **49/50** ?

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